

To adequately reinforce learning, Zeenat plays the same video for a week, and assesses them on a Saturday. Over the school year, the word list will increase in difficulty and the kids will also be provided with more complex phonic rules and word/sentence decoding strategies. Zeenat also sets aside time at the end of the day, when a student gets individual time on her laptop, to work on these skills using a phonics improvement software. Each child is shown either a visual and/or the spelling of a word and s/he then has to speak the word aloud into an attached speaker. The voice processing software graphically shows which syllables the child is able to pronounce correctly; it also visually highlights the incorrectly read syllables and speaks them out again.

Maths and Science

Each math and science lesson on a new topic begins with a short video sourced from Khan Academy (<https://www.khan-academy.org>) and other websites. Zeenat first plays the video in its entirety, and then for a second time while pausing it at critical points and reinforcing the concepts that are being discussed. A part of the screen is dedicated to the keypoints for



Interactive maths learnt via "Kinect Sesame Street"

the topic, and remains visible throughout the hour-long lesson. Students can keep referring back to them when they start solving problems independently. At the end of every lesson, Zeenat gives them assignments/quizzes to attempt from home on their TVs 4 Learning set-top box.

Reinventing computer labs

Soon, it's time for computer class, and the kids go down to a small room with 10 aged PCs that functions as the school's computer lab, which they get to attend one hour every week. At least three students sit together at the same workstation and keep shuffling the mouse and keyboard

EXPERTSPEAK



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What will a future classroom, with optimum technology penetration, look like (and will we even need a classroom)?

All students excel when teachers, as instructional experts, can leverage technology (for both basic skills instruction and data collection) in a really meaningful way while also being able to cater to the various other necessary modalities that are required for high student achievement. Physically, this optimized learning space includes areas for computer use, whole-class instruction from teachers, independent exploration, tablets, and small group remediation led by tutors in addition to various other opportunities as well.

Can technology ever completely replace the human teacher?

No. Technology is a great complement to learning and instruction, however, the interaction that is unique between teachers and students, the learning that occurs simply from interaction, and the higher level content that can only be taught by people, is critical to learning and ensuring that students are well-rounded.

among them. However, Adithya and his two best friends Zafar and Prakriti have long since learnt "keyboard racing", and they place non-monetary bets on who can type the fastest. To avoid any disputes, and for that added kick, they stick to www.app.typrx.com – a worldwide typing game! Adithya gets a couple of quick pinches on the shoulder as his deviates from a touch-typing position. (Their computer teacher is really trying to get her students to stop being slaves to their two index fingers when on the keyboard.)

Zeenat has managed to convince her school management to let the lab be used for more than its intended purpose i.e "learning how to use a computer". Each weekly session is dedicated to a specific subject (languages, maths, science, and social studies), and the kids are required to use the computer as a tool to deepen their understand of what they have learnt in each subject over the unit. Today's session is dedicated to the languages, and as Adithya, Zafar and Prakriti get started with their emails to Peter Jackson, they come across points where they cannot directly express their ideas in the English language. At such moments, they narrate it to a voice interpretation and translation software, which gives them three variants in English. For a session focussed on maths, Zeenat asks her students to attempt interactive quizzes (which significantly promote higher-order thinking; see Bloom's Taxonomy at <http://dgit.in/YAyV3U>) on online resources such as www.MyiMaths.com. To promote interest in the sciences and social sciences, the kids are asked to view pre-downloaded YouTube videos on topics as diverse as The Ice Age to The Aborigines in Australia!

Stitching it all together

In any foreseeable future, while technology may not completely replace a human teacher in a school classroom, it has the potential to make giant leaps on several fronts. As Preston Smith Co-Founder and CEO of Rocketship Foundation, (see box "Masterclass") says, "Online learning programs are really great at teaching basic skills and providing opportunities for guided and independent practice. Our teachers, on the other hand teach higher-order skills, like critical thinking, sharing, and